Qatar University

College of Engineering

Dept. of Computer Science & Eng.

Software Engineering

CMPS 310 - Lab

Lab 8: Design Sequence Diagram

Supporting tool - Visual Paradigm | 2 hours 30 minutes

Learning objectives

- Develop design sequence diagram (DSD) of the system.
- Understanding of Object collaboration in terms of message passing.
- Understanding of Ordering of message
- Understanding of Concepts of collection (container/database class)

Course learning outcomes: CLO 2 - Elicit, analyze, prioritize and document requirements for a software system. CLO 6 - Design, implement, test and present a software system in a group.

Class Diagram

A sequence diagram is used primarily to show the interactions between objects that are represented as lifelines in a sequential order. This lab helps you prepare the sequence diagram required for Milestone 2 of your course project.

Use Visual Paradigm to create your class diagram. Add Actor, Lifeline, Boundary Object, Control Object, Entity Object, Collection / Container, Synchronous Message, Return Message, Self Message, Activation Bar, Alternative Fragment (alt), Optional Fragment (opt), Loop Fragment, Object Creation, Time Ordering using the notation below.

Name	Symbol	Meaning and direction
Actor		Represents an external user or system that interacts with the system.
Lifeline	ObjectName	Represents an object participating in the interaction. The dashed vertical line shows its existence over time.
Boundary Object	«boundary»	Represents the interface between an actor and the system.
Control Object	«control»	Coordinates system operations and controls the flow of messages.
Entity Object	«entity»	Represents an object that stores or manages system data.

Name	Symbol	Meaning and direction
Collection / Container	ClientContainer	Represents a collection of objects used to store, find, or retrieve information.
Synchronous Message	——————▶	Represents a method call from one object to another. The arrow points to the receiving object.
Return Message	·····▷	Represents a response or returned value. The dashed arrow points back to the sender.
Self Message	↳	Represents an object calling one of its own methods.
Activation Bar	▮	Shows the period during which an object is performing an operation.
Alternative Fragment (alt)	alt [condition]	Represents alternative execution paths based on conditions, such as client exists / does not exist.
Optional Fragment (opt)	opt [condition]	Represents an interaction that executes only when a condition is true.
Loop Fragment	loop [condition]	Represents messages that are repeated while a condition is satisfied.
Object Creation	«create» ——▶	Represents the creation of a new object during an interaction.
Time Ordering	↓	Messages are read from top to bottom to show the chronological order of interactions.

Part A QU Print Shop - Guided practice

A.1 Identify the actor, system response, object, message between objects from the use case specification

Recall the Print Shop: users submit and revise queued jobs, printing runs automatically, and collection may include a receipt. PAS handles credit and charges; staff can request summaries of their own jobs.

Start from the Print Shop use case diagram in Lab 5, the worked specifications of UC001 Submit print job and UC003 Collect printed document in Lab 6 and class diagram in Lab 7

The worked table below identifies the actor, system response, object, message between objects from the use case specification from the use case specification and supporting evidence

UC001 – Submit print job

Step	Actor action	System response	Responsible object	Message between objects	Supporting evidence
1	User uploads a doc-	Receives the document	PrintShop	submitJob(user, doc,	Lab 6, UC001, Step 1; Lab

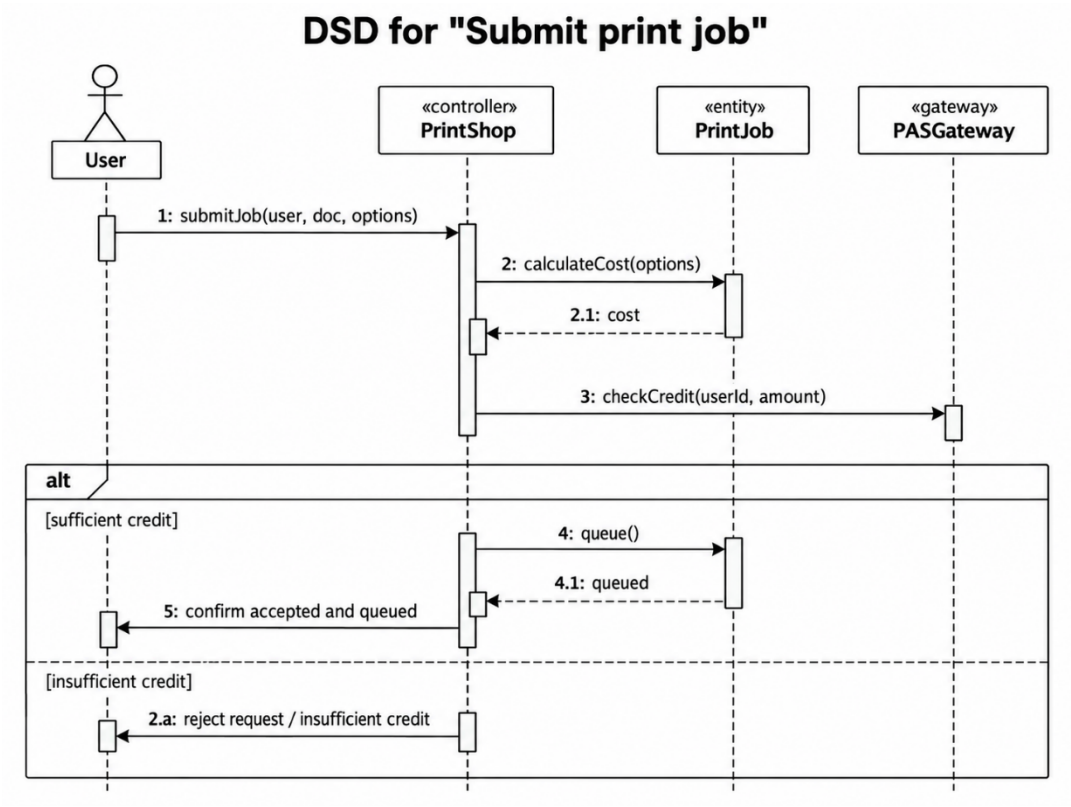
	ument and selects printing options.	and options.		options)	7, PrintShop operations.
2	—	Calculates the printing cost.	PrintJob	calculateCost(options)	Lab 6, UC001, Step 2; Lab 7, PrintJob operations.
3	—	Requests a credit check from PAS.	PASGateway	checkCredit(userId, amount)	Lab 6, UC001, Step 2; Lab 7, PASGateway operations.
4	—	Records the accepted job and sets its status to QUEUED.	PrintJob	queue()	Lab 6, UC001, Step 3; Lab 7, PrintJob operations.
5	—	Confirms that the job has been accepted and queued.	PrintShop	Return confirmation to User	Lab 6, UC001, Step 4.
2.a	—	If credit is insufficient, rejects the request and informs the user.	PrintShop	Return rejection to User	Lab 6, UC001, Alternative Flow 2.a.

UC003 – Collect printed document

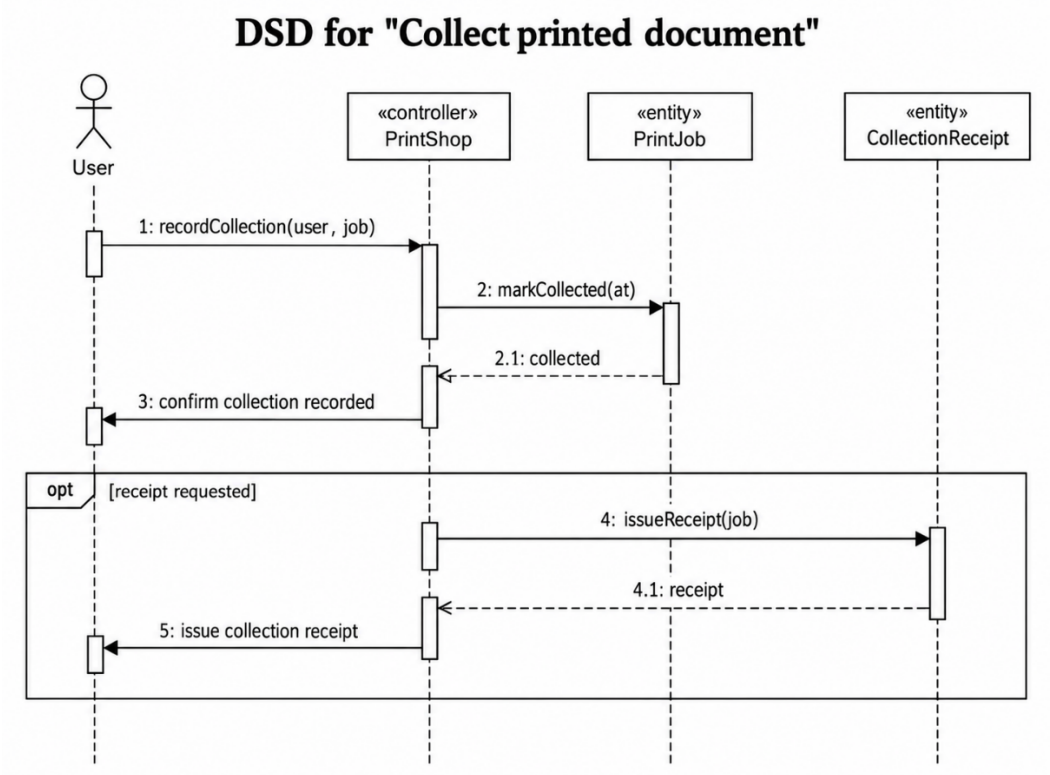
Step	Actor action	System response	Responsible object	Message between objects	Supporting evidence
1	User submits a collection request for a printed job.	Receives the collection request.	PrintShop	recordCollection(user, job)	Lab 6, UC003, Step 1; Lab 7, PrintShop operations.
2	—	Records collection and changes the job status to COLLECTED.	PrintJob	markCollected(at)	Lab 6, UC003, Step 2; Lab 7, PrintJob operations.
3	—	Confirms that collection has been recorded.	PrintShop	Return confirmation to User	Lab 6, UC003, Step 3.
Extension	User requests a collection receipt (optional).	Issues a receipt after collection is recorded.	PrintShop	issueReceipt(job)	Lab 5, UC004; Lab 6, UC003 extension point; Lab 7, PrintShop operations.

A.2 Design sequence diagram

Submit print job:



Collect Printed document



Part B All Phones Shop

Individual assessment. Begin with a 10-minute instructor review of the Lab 6 APS diagram and Lab 7 APS Reference Specifications: UC001 Register client, UC003 List item for sale and UC006 Place bid.

B.1 Identify the actor, system response, object, message between objects from the use case specification - 25 minutes

Use the APS case from Lab 3 and Lab 6, Figure 2 and the Register client example. Keep PoB financial accounts and login outside APS.

Identify the actor, system response, object, message between objects from the use case specification from the use case specification and supporting evidence

Step	Actor action	System response	Responsible object	Message between objects	Supporting evidence

B.2 Design sequence diagram

Develop the diagram in Visual Paradigm using B.1. Show the actor, system response, object, message between objects from the use case specification from the use case specification and supporting evidence .

B.3 Assign operations from the use case specifications - 30 minutes

Derive operations from UC001 Register client, UC003 List item for sale and UC006 Place bid. For each, record its responsible class, typed parameters, return type and source use case step. Add the operations to your diagram.

Class	Operation with parameter and return types	Use case step and placement reason

B.3 Review and refine the APS sequence diagram - 15 minutes

Exchange diagrams with a classmate. the actor, system response, object, message between objects from the use case specification from the use case specification and supporting evidence. Record one strength, one suggested improvement and the change you made.

Submission requirements

Allow 15 minutes. Submit the three files below in a folder named **Lab 8 Sequence Diagram** in your own repository in the CMPS310 course organization. Follow the commit and push workflow from Lab 2, then check that the files open on GitHub. Part A is guided practice and is not submitted.

File	Required contents
lab08-solution.docx	Name and student ID; completed Part B tables; final APS sequence diagram.
aps-sequence-diagram.png	Readable export of the final APS sequence diagram.
aps-sequence-diagram.vpp	Editable Visual Paradigm project containing the APS sequence diagram.

Assessment criteria (10 points)

Criterion	Points
Identify the actor, system response, object, message between objects from the use case specification (UC001)	2.5
Identify the actor, system response, object, message between objects from the use case specification (UC003)	2.5
Design sequence diagram (UC001)	2
Design sequence diagram (UC003)	2
Review and refine the APS sequence diagram -	1

Further reading

Visual Paradigm - What is sequence Diagram?